CHAT HISTORY

MAZE RUNNER

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User: /init

GitHub Copilot: Optimized tool selection

User: Read my README.md and inspect my existing Maze Chase Pygame project before changing anything.

Implement Task 1 only: Multiple Enemies.

Currently, one enemy uses BFS pathfinding to chase the player. Add two more enemies at different valid maze corners, so there are three enemies total.

Requirements:

- Reuse the existing Enemy class and BFS pathfinding logic.
- Each enemy must independently chase the player.
- Update and draw all three enemies.
- Check collisions with every enemy.
- Ensure enemies spawn in walkable cells and do not overlap the player.
- Preserve existing controls, maze generation, and restart functionality.
- Do not implement Tasks 2, 3, or 4 yet.
- Explain which files you changed and how I can test the feature.

Make changes directly to my existing project. Do not create a new game or overwrite unrelated code.

GitHub Copilot: I'm reviewing the existing project and README first so I can make the smallest safe change for Task 1 and keep the current game behavior intact.

Read []()

Searched for text `Enemy|BFS|enemy|pathfind|maze|restart|collision` (`**/*.py`), 17 results

Read []()

Read []()

Read []()

Read []()

I’ve confirmed the current game only creates one enemy and checks a single collision. I’m changing just that flow to a list of three independent enemy instances while keeping the original restart and movement logic intact.

Read []()

Ran terminal command: cd c:/Users/keert/11_maze_chase ; $env:SDL_VIDEODRIVER='dummy'; python -c "from game.game_engine import GameEngine; g=GameEngine(); print('has_enemies', hasattr(g, 'enemies')); print('enemy_type', type(getattr(g,'enemy',None)).__name__ if hasattr(g,'enemy') else 'missing')"

Read []()

Replacing 25 lines with 32 lines in []()

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The change is in place; I’m validating the project still boots with the expected three-enemy setup using a quick smoke test in the existing codebase.

Ran terminal command: cd c:/Users/keert/11_maze_chase ; python -m pip install -r requirements.txt

Ran terminal command: cd c:/Users/keert/11_maze_chase ; $env:SDL_VIDEODRIVER='dummy'; python - <<'PY'

from game.game_engine import GameEngine

g = GameEngine()

print('enemy_count', len(g.enemies))

print('positions', [(e.r, e.c) for e in g.enemies])

print('all_valid_spawn', all((0 <= e.r < 11 and 0 <= e.c < 13) for e in g.enemies))

print('player_overlap', any(g.player.rect.colliderect(e.rect) for e in g.enemies))

print('first_enemy_is_list_member', g.enemy is g.enemies[0])

PY

Ran terminal command: cd c:/Users/keert/11_maze_chase; $env:SDL_VIDEODRIVER='dummy'; python -c "from game.game_engine import GameEngine; g = GameEngine(); print('enemy_count', len(g.enemies)); print('positions', [(e.r, e.c) for e in g.enemies]); print('player_overlap', any(g.player.rect.colliderect(e.rect) for e in g.enemies)); print('first_enemy_alias', g.enemy is g.enemies[0]); print('valid_cells', all(0 <= e.r < 11 and 0 <= e.c < 13 for e in g.enemies));"

The first smoke test tripped over PowerShell quoting rather than the game logic, so I'm re-running the exact same verification with a shell-safe Python one-liner.

Ran terminal command: cd c:/Users/keert/11_maze_chase; $env:SDL_VIDEODRIVER='dummy'; python -c "from game.game_engine import GameEngine; g=GameEngine(); print('enemy_count', len(g.enemies)); print([(e.r,e.c) for e in g.enemies]); print('player_overlap', any(g.player.rect.colliderect(e.rect) for e in g.enemies)); print('alias_ok', g.enemy is g.enemies[0]); print('valid_spawn', all(0 <= e.r < 11 and 0 <= e.c < 13 for e in g.enemies));"

Completed with input: cd c:/Users/keert/11_maze_chase; $env:SDL_VIDEODRIVER='dummy'; python -c "from game.game_engine import GameEngine; g=GameEngine(); print('enemy_count', len(g.enemies)); print([(e.r,e.c) for e in g.enemies]); print('player_overlap', any(g.player.rect.colliderect(e.rect) for e in g.enemies)); print('alias_ok', g.enemy is g.enemies[0]); print('valid_spawn', all(0 <= e.r < 11 and 0 <= e.c < 13 for e in g.enemies));"

Tool execution failed with exit code 1

Ran terminal command: cd "c:/Users/keert/11_maze_chase"; @'

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from game.game_engine import GameEngine
g = GameEngine()
print('enemy_count', len(g.enemies))
print([(e.r, e.c) for e in g.enemies])
print('player_overlap', any(g.player.rect.colliderect(e.rect) for e in g.enemies))
print('alias_ok', g.enemy is g.enemies[0])
print('valid_spawn', all((0 <= e.r < 11 and 0 <= e.c < 13) for e in g.enemies))
'@ | python -
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Ran terminal command: print(123)

Ran terminal command: print('ok'); from game.game_engine import GameEngine; g=GameEngine(); print('enemy_count=', len(g.enemies)); print('spawn_positions=', [(e.r, e.c) for e in g.enemies]); print('player_overlap=', any(g.player.rect.colliderect(e.rect) for e in g.enemies)); print('alias_ok=', g.enemy is g.enemies[0]); print('valid_spawn=', all((0 <= e.r < 11 and 0 <= e.c < 13) for e in g.enemies));

Ran terminal command: python

Ran terminal command: cmd /c "cd /d c:\Users\keert\11_maze_chase && python -c "from game.game_engine import GameEngine; g=GameEngine(); print('enemy_count=', len(g.enemies)); print('spawn_positions=', [(e.r, e.c) for e in g.enemies]); print('player_overlap=', any(g.player.rect.colliderect(e.rect) for e in g.enemies)); print('alias_ok=', g.enemy is g.enemies[0]); print('valid_spawn=', all((0 <= e.r < 11 and 0 <= e.c < 13) for e in g.enemies))""

Made changes.

User: Implement Task 2 from README.md only: Speed Up Over Time.

Every 15 seconds of gameplay, decrease the enemy movement interval by 2, making enemies move faster. Enforce a minimum movement interval of 5 so enemies never become too fast.

Requirements:

- Inspect the current code and preserve Task 1's three independent enemies.

- Use pygame.time.get_ticks() or equivalent elapsed-time tracking.

- Apply the speed increase consistently to all enemies.

- Avoid reducing the interval multiple times within the same 15-second period.

- Reset the difficulty timer and speed when the game restarts.

- Do not implement Tasks 3 or 4 yet.

- Explain which files you changed and how I should test the feature.

Modify the existing project directly and preserve all current functionality.

GitHub Copilot: I've checked the current project state and I'm adding the difficulty ramp without disturbing the three-enemy setup from Task 1.

Replacing 32 lines with 52 lines in []()

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Read []()

Ran terminal command: $env:SDL_VIDEODRIVER='dummy'; python -c "import pygame; from game.game_engine import GameEngine; g = GameEngine(); print('initial', [e.move_interval for e in g.enemies], g.speed_tier); g.last_speed_update = pygame.time.get_ticks() - 15000; g._update_enemy_speed(); print('after_15s', [e.move_interval for e in g.enemies], g.speed_tier); g.reset(); print('after_reset', [e.move_interval for e in g.enemies], g.speed_tier);"

Ran terminal command: cmd /c "cd /d c:\Users\keert\11_maze_chase && python -c "import sys; print(sys.executable); print('hello')""

Ran terminal command: cmd /c "cd /d c:\Users\keert\11_maze_chase && python -m py_compile main.py game\entities.py game\game_engine.py game\maze.py"

Made changes.

User: Fix the overlapping HUD text in my existing Maze Chase Pygame game. The speed/difficulty indicator for Task 2 overlaps the original "Reach EXIT before the enemy catches you!" message at the bottom. Inspect the existing drawing code and adjust the layout so both labels are clearly readable, with suitable spacing or separate positions. Preserve the maze, all three enemies, movement, and difficulty logic. Do not implement Tasks 3 or 4 yet. Make the smallest necessary change and explain which file you modified.

GitHub Copilot: I'm adjusting just the HUD layout in the existing game so the speed tier and the original status text sit in separate, readable rows without changing the gameplay logic.

Read []()

Replacing 8 lines with 8 lines in []()

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Ran terminal command: cd "c:/Users/keert/11_maze_chase" ; python -m py_compile main.py game/entities.py game/game_engine.py game/maze.py

✅ HUD overlap fix

I fixed the overlap in the bottom HUD by changing the layout in `game_engine.py`.

What changed:

- The bottom panel height was increased from 50 to 60 pixels.

- The “Speed Tier” label now renders on the first line.

- The original “Reach EXIT before the enemy catches you! R=Restart” text now renders on the second line.

- This keeps the same gameplay logic, three-enemy behavior, and difficulty timer intact while making both labels readable.

Updated file

- `game_engine.py`

How to test

1. Open the project in your terminal.

2. Run:

   python `main.py`

3. Play the game and watch the bottom HUD.

4. Confirm:

   - “Speed Tier: X” appears on one line

   - “Reach EXIT before the enemy catches you! R=Restart” appears on a separate line

   - All three enemies, maze movement, and difficulty logic still work normally

Verification

I validated the project still compiles successfully with:

- python -m py_compile `main.py` `entities.py` `game_engine.py` `maze.py`

This completed without syntax errors.

Made changes.

User: Implement Task 3 from README.md only: Power Pellet (Freeze Enemy).

Add a yellow circular power pellet somewhere in a valid, walkable maze cell. When the player collects it, freeze all enemies for exactly 5 seconds.

Requirements:

- Inspect the existing maze, player, and enemy code before editing.

- Draw the pellet clearly in yellow.

- Detect when the player reaches the pellet.

- Freeze all three enemies for 5 seconds using pygame.time.get_ticks() or equivalent elapsed-time tracking.

- Enemies must not move while frozen and must resume chasing after 5 seconds.

- Display a clear visual indicator while enemies are frozen.

- Prevent the pellet from being collected repeatedly.

- Reset the pellet and freeze state when the game restarts.

- Preserve Tasks 1 and 2, including multiple enemies and increasing difficulty.

- Do not implement Task 4 yet.

- Explain the files changed and how to test the feature.

Make changes directly to the existing project. Do not replace the game or modify unrelated functionality.

GitHub Copilot: I'm reviewing the current movement and reset flow before adding the pellet so the freeze mechanic fits the existing three-enemy difficulty ramp without disturbing anything else.

Read []()

Read

Read []()

Replacing 82 lines with 124 lines in [](),

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Read []()

✅ Task 3 implemented: power pellet freeze

I updated the existing game logic in:

- `game_engine.py`

- `entities.py`

What changed

- Added a yellow power pellet at a valid, walkable maze cell.

- The pellet is collected once and then deactivates so it cannot be picked up repeatedly.

- When the player touches the pellet, all three enemies freeze for exactly 5 seconds using `pygame.time.get_ticks()`.

- Enemy movement is skipped while frozen, and they resume chasing normally after the timer ends.

- Added a clear visual freeze indicator:

  - frozen enemies get a blue outline

  - HUD shows `FROZEN! 5s`

- Reset behavior clears the pellet and freeze timer so a new round starts cleanly.

- The existing multiple-enemy behavior and speed ramp from Tasks 1 and 2 remain intact.

Files modified

- `game_engine.py`

- `entities.py`

How to test

1. Start the game:

   python `main.py`

2. Move the player around the maze until you find the yellow pellet.

3. Collect it.

4. Confirm:

   - the pellet disappears after collection

   - all enemies stop moving for about 5 seconds

   - enemies resume movement afterward

   - the frozen indicator appears while the effect is active

   - restarting with `R` resets the pellet and freeze state

Verification

I verified the edited project still compiles without syntax errors using:

- python -m py_compile `main.py` `entities.py` `game_engine.py` `maze.py`

This completed successfully with no reported syntax errors.

Made changes.

User: Implement Task 4 from README.md only: Score by Distance (survival score).

Requirements:

- Add a survival score to the existing Maze Chase game.

- Increment self.score by 1 every active gameplay frame while the player is alive.

- Display the survival score in the HUD.

- Show the survival time in seconds using score // 60, as specified in the README.

- Display the final score on the game-over screen.

- Do not increase the score after game over.

- Reset the score correctly when the player restarts using R.

- Preserve Tasks 1, 2, and 3: three independent enemies, progressive enemy speed, and the freeze power pellet.

- Fix any HUD overlap caused by adding the score.

- Modify the existing project directly; do not create a new game.

- Explain the files changed and how to test the feature.

GitHub Copilot: I'm keeping the existing pellet and enemy-speed mechanics intact and adding only the survival-score tracking and HUD adjustments needed for Task 4.

Read []()

Replacing 64 lines with 70 lines in []()

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Ran terminal command: cd "c:/Users/keert/11_maze_chase" ; python -m py_compile main.py game/entities.py game/game_engine.py game/maze.py

✅ Task 4 implemented: survival score

I updated the existing game in `game_engine.py` to add the survival score and keep the working Tasks 1–3 intact.

What changed

- Added `self.score` to the game state and reset it on restart.

- Increased the score by 1 on every active gameplay frame while the player is alive.

- Showed the score in the HUD as:

  - `Survived: Xs`

  - where `X = score // 60`

- Displayed the final survival time on the end-of-game overlay.

- Prevented score growth after game over.

- Kept the existing three-enemy system, enemy speed ramp, and power-pellet freeze mechanic unchanged.

- Adjusted the HUD layout so the new score text does not overlap with the existing status text.

Files changed

- `game_engine.py`

How to test

1. Run:

   python `main.py`

2. Move around the maze and stay alive.

3. Watch the bottom HUD update with the survival time.

4. Let the enemies catch you or reach the exit.

5. Confirm:

   - the score increases every frame while alive

   - the HUD shows the elapsed seconds as `Survived: Ns`

   - the final score appears on the game-over overlay

   - pressing `R` resets the score correctly

Verification

I validated the project still compiles after the change with:

- `python -m py_compile `main.py` `entities.py` `game_engine.py` game/maze.py`

This completed without syntax errors.

Made changes.